

Semiconductor, Honeywell, Delphi, Huawei, Tata Elxsi, Mistral, LG Electronics, Samsung, Microchip Technology, eInfochips, Serveen, ATS, Sankhya Technologies, Bluefont, Intoto, Wind River and DCM Technologies.

"In the embedded industry, the jobs exist in the early phases of product life cycle like design, verification and validation, architecture support, functional test and rapid control prototyping," adds Solaikutty Dhanbal, manager, NI India Academics.

Marc McComb, academic program sales engineer, Microchip Technology Inc., elaborates, "As new technologies emerge, embedded systems are being integrated into more products. Therefore engineering students with knowledge related to embedded systems will find that their skills will become increasingly valuable to a number of different sectors including, but not limited to, automotive, medical, consumer electronics, aerospace, military, and much more. Also, the current trend in the industry seems to be expanding the applications that utilise embedded systems. With India's wealth of engineering talent, I believe manufacturers will take advantage of this and continue to hire from this region."

"The opportunities for embedded



Microchip looks for certain skills on a case-by-case basis, for a particular focus. For example, some applications are more software-centric, such as our touch-sensing technologies. In this case, a priority could be placed on a candidate's abilities to create efficient and effective firmware, while still being able to understand how their software will interact with the rest of the system — Marc McComb, academic program sales engineer, Microchip Technology Inc.

professionals exist in product-related companies, software companies which do design for product companies elsewhere and research field as well," Kumaran informs.

He points out, "There is a need of embedded engineers across verticals in R&D, automotive, telecom infrastructure to engineer the LTE devices, base stations etc. Apart from these, embedded engineers are much in demand in the consumer space as well as in the gaming industry. Android is one of the fastest growing platforms so there is a plenty of scope for embedded engineers here too."

Newer tools that make it easier to utilise embedded systems have definitely created new markets. Such new markets include energy harvesting,

lighting, intelligent power systems, human interface such as touch-screen controllers, and many more.

How to get the entry pass?

Let's delve a little deeper to know the basic criteria for entry. To work in this field, diploma holders, engineering (as well as science) graduates, post-graduates and even doctorates with the background of electronics/electrical engineering or computer science may try their luck here.

Now coming to entry-level recruitment, it demands some pre-defined skillsets. On the software side, it is mandatory to have the basic knowledge of 'C' and assembly language programming. On the other side, sound knowledge of basic electronics, especially digital and analogue design techniques, microcontrollers and microprocessors, and system knowledge related to communication protocol and real-time operating system can be helpful in gaining a strong foothold in the field.

This may be one of the fastest growing sectors of electronics, but there is a scarcity of people with knowledge of embedded system design. In fact, it is a challenge to find a fresher with a good grasp of the basic electronics and designing concepts.

What you can get?

Wondering whether this field is powerful enough to fill your pocket to a good extent? Well, it depends primarily on the nature of organisation,

